Side project · open method · MIT

SMETP

Subject-Matter Expertise Transfer Protocol

Augment your experts. Don't replace them.

SMETP — capturing expert judgment so AI can follow it and show its work

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The slide every AI deck skips

Who actually knows how this decision gets made?





THE MISSING INPUT

Most AI plans start with the model and end with a prompt. The expert never makes it in.

The question I tested

If you write an expert's judgment down formally, what do you gain over just asking an LLM?

8
inputs

26
decision-tree nodes

116
knowledge-graph triples



CREDIT-RISK EXAMPLE

Fictional senior credit analyst



TESTABLE GRAPH

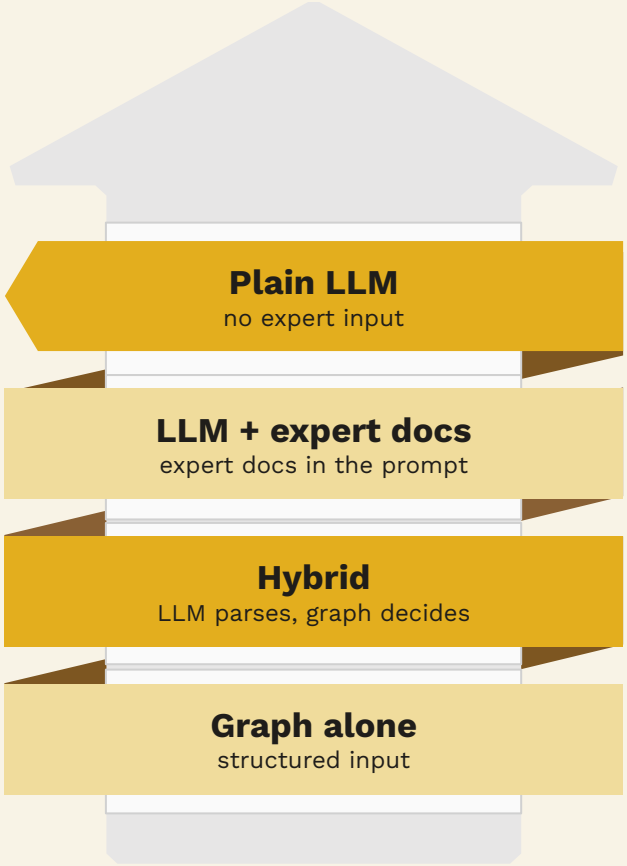
Validator + deterministic reasoner

The setup

Four systems, one expert's rules

139 synthetic cases

80 in-distribution · 16 boundary
3 doc-conflict · 24 out-of-scope
16 adversarial



Without the expert

30.2%

DECISION ACCURACY · PLAIN LLM

No expert rules. Safe-ish, mostly useless.

It escalated almost everything.

58 of 97

cases gave the same answer across 3 runs

The results

Hybrid: same accuracy, a sixth of the cost

Setup	Decision accuracy	Rule citation	Median latency	Cost (139)
Plain LLM	30.2%	3.6%	8.7 s	$0.87
LLM + expert docs	97.1%	90.6%	16.8 s	$2.29
Hybrid: LLM parses, graph decides	**97.1%**	**100%**	**2.5 s**	**$0.36**
Graph alone (structured input)	100%*	100%	~0.008 ms	$0

*By construction — test answers come from the same rules the graph encodes.

Synthetic credit-risk benchmark · 139 cases · grok-4.7 · 1 run per case · Oct 2026

Cost and latency

About a sixth of the cost, and faster

DOCS IN PROMPT

16.8 s

median · $2.29 for 139

~6,000 tokens per decision

HYBRID

2.5 s

median · $0.36 for 139

~1,644 tokens per decision

THE POINT

~1/6

of the cost

Hybrid citations come from the rule path, not model-written prose

Synthetic credit-risk benchmark · 139 cases · grok-4.7 · 1 run per case · Oct 2026

Read the 100% correctly

A fidelity check, not “graph beats LLM”

THE QUESTION TO ASK

If someone sells “graph beats LLM” from a setup like this, ask where the ground truth came from.

WHAT IT SHOWS

Agreement with an independent oracle

GRAPH ALONE · DECISION ACCURACY

100%*

*By construction — the test answers come from the same rules the graph encodes.

Synthetic credit-risk benchmark · 139 cases · grok-4.7 · 1 run per case · Oct 2026

The hybrid architecture

The LLM reads. The graph decides.

TESTED ON

Free-text notes

INTENDED, NOT YET TESTED

Emails and PDFs



1 · LLM EXTRACTS

Pulls fields from free text. Told to omit, never guess.



2 · SCHEMA GUARD

Required fields, types, ranges. Notes text never reaches the decider.



3 · GRAPH DECIDES

Deterministic walk, with the rule path. Unknowns escalate to a human, with the reason.

ONE MISS, TRACED

credit_score: "800"

a string, not a number

→ parser “helpfully” converts it
→ graph approves
→ should have escalated

THE CAUSE

Raw types were not checked before the parser ran

Where it failed

All 4 hybrid misses, one cause.

01 The same 4 cases drive the hybrid's 12.9% overreach on must-escalate cases

02 Fix: validate raw types before the parser, or have the parser report mismatches

03 Rerun with the fix is first on the roadmap, not done yet

04 Adversarial 87.5% vs 75% (14 vs 12 of 16): within noise

Synthetic credit-risk benchmark · 139 cases · grok-4.7 · 1 run per case · Oct 2026

What encoding the tree found

The best result wasn't a score

2 of 5 — published test cases contradict the decision tree

650+ — bankruptcy was never checked at a credit score of 650 or higher

THE POINT

Prose hid all of this. A graph can be tested.

ALSO FOUND

Fields used by rules but never declared

Reason text that disagrees with the logic

DENY / ESCALATE outcomes with no confidence

THE METHOD

01 Capture
02 Analyze
03 Codify
04 Validate
05 Deploy

What SMETP is

An open method for capturing expert judgment

WHAT GETS CAPTURED

Inputs · decision tree · outcomes + confidence · heuristics · edge cases · validation cases

INTERVIEW

Structured, role-specific expert interview

OPEN, MIT

Method docs and templates on GitHub

3 EXAMPLES

Worked: finance, healthcare, legal

An open method with worked examples — not shipped packages.

Open today vs roadmap

What's open today, and what isn't yet

No npm packages or drop-in adapters yet.

OPEN TODAY

Free on GitHub, MIT

Method: FRAMEWORK.md, GETTING_STARTED.md
Interview templates
3 JSON examples: finance, healthcare, legal
Oct 2026 credit-risk benchmark report

ROADMAP

Not shipped. Every item below is planned.

- **ROADMAP** Public benchmark harness
- **ROADMAP** JSON Schema + validator
- **ROADMAP** SKILL.md compile
- **ROADMAP** MCP server (decide / explain / validate)
- **ROADMAP** Healthcare + legal benchmarks
- **ROADMAP** Privacy / consent model

The free path

Start with the method, not a sign-up

GITHUB-FIRST

Open method, MIT. Read it, run the examples, tell me what breaks.

01 THE METHOD
GETTING_STARTED.md + templates

02 WORKED EXAMPLES
Finance, healthcare, legal (JSON)

03 YOUR DECISION
Hate to lose it? Open an issue

START HERE

SMETP on GitHub

github.com/goker/SMETP-Subject-Matter-Expertise-Transfer-Protocol

THE RULE I'M HOLDING TO

If the code doesn't show it, the slide doesn't say it.

Limits, stated plainly

What I won't claim yet

01 No published @smetp/* npm packages

02 No drop-in Claude / ChatGPT / Gemini adapters

03 No accuracy figures from before this benchmark

04 Graph-alone 100%* is by construction, not proof a graph beats an LLM

WHAT I DO CLAIM

Hybrid: 97.1% with 100% rule citations on 139 synthetic cases

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Thank you · one ask

Which expert decision should we capture next?

START HERE
github.com/goker/SMETP-Subject-Matter-Expertise-Transfer-Protocol

BRING A DECISION
Open an issue or Discussion: one you'd hate to lose

REPLICATE IT
Builders: rerun it, break it, open issues

Appendix A · method detail

How the method works

BACKUP FOR Q&A

Capture → Codify → Hybrid

Hybrid tested on free-text notes only



1 · CAPTURE

Structured, role-specific expert interview, with templates.



2 · CODIFY

Decision tree → testable graph, checked by a validator.



3 · HYBRID

The LLM reads, the rules decide, unknowns escalate to a person.

Appendix B · full results

Oct 2026 benchmark, all four setups

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Synthetic data, one domain, one model, one run per case. Limitations in Appendix C.

Synthetic credit-risk benchmark · 139 cases · grok-4.7 · 1 run per case · Oct 2026

ABOUT THE DATA

Synthetic cases, not real applications

One domain: credit risk

16 adversarial cases (8 scenarios × 2)

Tree treated as authoritative where docs conflict

NOT YET PUBLIC

The benchmark harness. Publishing it is on the roadmap.

Appendix C · limitations

What this benchmark doesn't show

01 One model (grok-4.7, temp 0.7), one scored run per case

02 Consistency measured only for the plain LLM

03 Lightweight Python KG, not full OWL/RDF

04 No privacy or PII testing